



警示

1. 实验报告如有雷同，雷同各方当次实验成绩均以 0 分计。
2. 当次小组成员成绩只计学号、姓名登录在下表中的。
3. 在规定时间内未上交实验报告的，不得以其他方式补交，当次成绩按 0 分计。
4. 实验报告文件以 PDF 格式提交。

院系	计算机学院	班 级	计科 1 班	组长	陈安康
学号	22320021	22318001	22336070	22336064	
学生	陈安康	蔡可忻	冯乙山		

编程实验

【实验内容】

1. 在实验设备上完成 P243 实验 7-2 并测试实验网连通性 (如果使用 packet tracer 注意是三层交换机)。
2. 通过实验观察 RIP V1 和 V2 的区别 (重点在 VLSM 上) 给出分析过程与结果 (实验 IP 采用 10.10.x.0 网段)
3. 学会使用 Debug ip packet 和 Debug ip rip 命令，并对 debug 信息做分析。
4. 观察试验拓扑中链路状态发生改变时路由表的前后信息对比及 debug 信息的变化。

【实验目的】

1. 学习 RIP 的基本配置，理解 RIP 的机制。区分 RIPv1 与 RIPv2 的不同。学会依据调试信息解决问题。

学号	学生	自评分
22320021	陈安康	100
22318001	蔡可忻	100
22336070	冯乙山	100
实验分工见个人心得		

【交实验报告】

上传实验报告：助教

截止日期（不迟于）：1 周之内

上传包括两个文件：

(1) 小组实验报告。上传文件名格式：小组号_编程实验.pdf (由组长负责上传)

例如：文件名“10_编程实验.pdf”表示第 10 组的编程实验报告

(2) 小组成员实验体会。每个同学单独交一份只填写了实验体会的实验报告。只需填写自己的学号和姓名。

文件名格式：小组号_学号_姓名_编程实验.pdf (由组员自行上传)

例如：文件名“10_05373092_张三_编程实验.pdf”表示第 10 组的编程实验报告。

注意：不要打包上传！



7_2 实验:

步骤 1:

(1) 配置实验中的网络拓扑图后, 两个主机直接通过 ping 命令尝试通信, 发现 ping 不通。

(2) 在路由器 1, 2 上执行 show ip route 命令, 路由表信息如下, 可以看到, 路由表只显示自己直接相连的网段信息:

```
20-RSR20-1(config)#show ip route

Codes: C - connected, S - static, R - RIP, B - BGP
        O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default

Gateway of last resort is no set
C    192.168.1.0/24 is directly connected, GigabitEthernet 0/1
C    192.168.1.1/32 is local host.
C    192.168.2.0/24 is directly connected, Serial 2/0
C    192.168.2.1/32 is local host.
```

```
20-RSR20-2(config)#show ip route

Codes: C - connected, S - static, R - RIP, B - BGP
        O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default

Gateway of last resort is no set
C    192.168.2.0/24 is directly connected, Serial 2/0
C    192.168.2.2/32 is local host.
C    192.168.3.0/24 is directly connected, GigabitEthernet 0/1
C    192.168.3.1/32 is local host.
```

步骤 7:

在经过配置后, 通过 show ip route 命令查看路由表如下:

路由表 1:

```
show ip route

Codes: C - connected, S - static, R - RIP, B - BGP
        O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default

Gateway of last resort is no set
C    192.168.1.0/24 is directly connected, GigabitEthernet 0/1
C    192.168.1.1/32 is local host.
C    192.168.2.0/24 is directly connected, Serial 2/0
C    192.168.2.1/32 is local host.
R    192.168.3.0/24 [120/1] via 192.168.2.2, 00:00:06, Serial 2/0
R    192.168.5.0/24 [120/1] via 192.168.1.2, 00:19:54, GigabitEthernet 0/1
```

路由表 2:

```
20-RSR20-2(config-router)#show ip route

Codes: C - connected, S - static, R - RIP, B - BGP
        O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default

Gateway of last resort is no set
R    192.168.1.0/24 [120/1] via 192.168.2.1, 00:00:29, Serial 2/0
C    192.168.2.0/24 is directly connected, Serial 2/0
C    192.168.2.2/32 is local host.
C    192.168.3.0/24 is directly connected, GigabitEthernet 0/1
C    192.168.3.1/32 is local host.
```



交换机的路由表:

```
20-S5750-1(config)#show ip route

Codes: C - connected, S - static, R - RIP, B - BGP
        O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default

Gateway of last resort is no set
C    192.168.1.0/24 is directly connected, VLAN 10
C    192.168.1.2/32 is local host.
R    192.168.2.0/24 [120/1] via 192.168.1.1, 00:00:37, VLAN 10
R    192.168.3.0/24 [120/2] via 192.168.1.1, 00:00:37, VLAN 10
C    192.168.5.0/24 is directly connected, VLAN 50
C    192.168.5.1/32 is local host.
20-S5750-1(config)#
```

可以看到经过 RIP 学习，路由表中已经包含了其他网段的路由信息（带标签项 R 的条目）。其中的 RIP 条目是通过和其他交换机和路由器交换路由信息来更新得到的。

步骤 8:

在配置好 RIP 后，测试连通性，网络连通。

(1) 比较此时的路由表与步骤 1 的路由表（均在上文展示），会发现路由表中不仅包含与自己直连的网段信息，还增加了由 RIP 过程学习到的其他网段的路由信息（R 条目）。说明经过 RIP 学习，更新了路由表，使得路由器可以将报文传输到与自己不直接相连的网络中，实现网络互通。

(2) 执行在 PC1 上执行 tracert 192.168.3.22，命令结果如下：

```
C:\Windows\system32>tracert 192.168.3.22

通过最多 30 个跃点跟踪
到 D52_11 [192.168.3.22] 的路由:

 1  <1 毫秒    <1 毫秒    <1 毫秒  192.168.5.1
 2  <1 毫秒    <1 毫秒    <1 毫秒  192.168.1.1
 3  43 ms      41 ms       41 ms    192.168.2.2
 4  46 ms      49 ms       49 ms    D52_11 [192.168.3.22]

跟踪完成。
```

可以看到，数据包经过交换机-路由器 1-路由器 2，最后到达 PC2，这与路由表中记录的转发规则一致。

(3)

No.	Time	Source	Destination	Protocol	Length	Info
47	12.625366	192.168.3.1	224.0.0.9	RIPv2	70	Request
75	13.617856	192.168.3.1	224.0.0.9	RIPv2	70	Request
84	14.617991	192.168.3.1	224.0.0.9	RIPv2	70	Request
91	16.608924	192.168.3.1	224.0.0.9	RIPv2	110	Response
131	46.609757	192.168.3.1	224.0.0.9	RIPv2	110	Response
203	76.610399	192.168.3.1	224.0.0.9	RIPv2	110	Response
271	93.620692	192.168.3.1	224.0.0.9	RIPv2	70	Request
293	94.620116	192.168.3.1	224.0.0.9	RIPv2	70	Request
296	94.940127	192.168.3.1	224.0.0.9	RIPv2	110	Response
303	95.620165	192.168.3.1	224.0.0.9	RIPv2	70	Request
489	124.940878	192.168.3.1	224.0.0.9	RIPv2	110	Response
573	154.941416	192.168.3.1	224.0.0.9	RIPv2	110	Response
646	184.942322	192.168.3.1	224.0.0.9	RIPv2	110	Response
697	214.943058	192.168.3.1	224.0.0.9	RIPv2	110	Response
713	244.943976	192.168.3.1	224.0.0.9	RIPv2	110	Response
730	274.948432	192.168.3.1	224.0.0.9	RIPv2	110	Response
734	304.945410	192.168.3.1	224.0.0.9	RIPv2	110	Response
748	334.945994	192.168.3.1	224.0.0.9	RIPv2	110	Response
752	364.946719	192.168.3.1	224.0.0.9	RIPv2	110	Response

拔掉网线后，路由数据传输变快

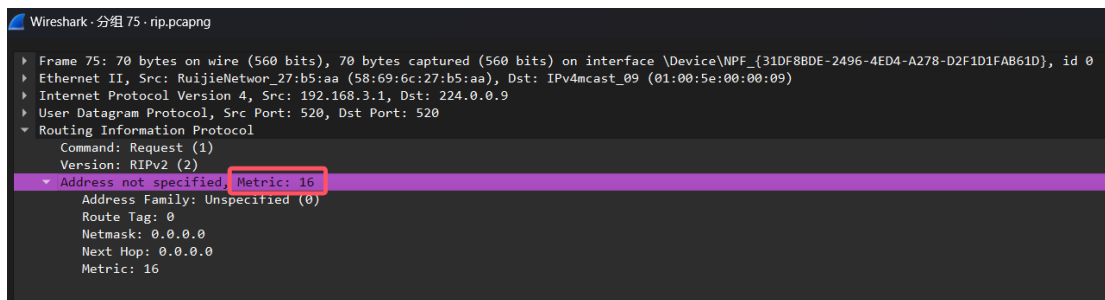
正常情况下，稳定每30秒传输一次路由信息



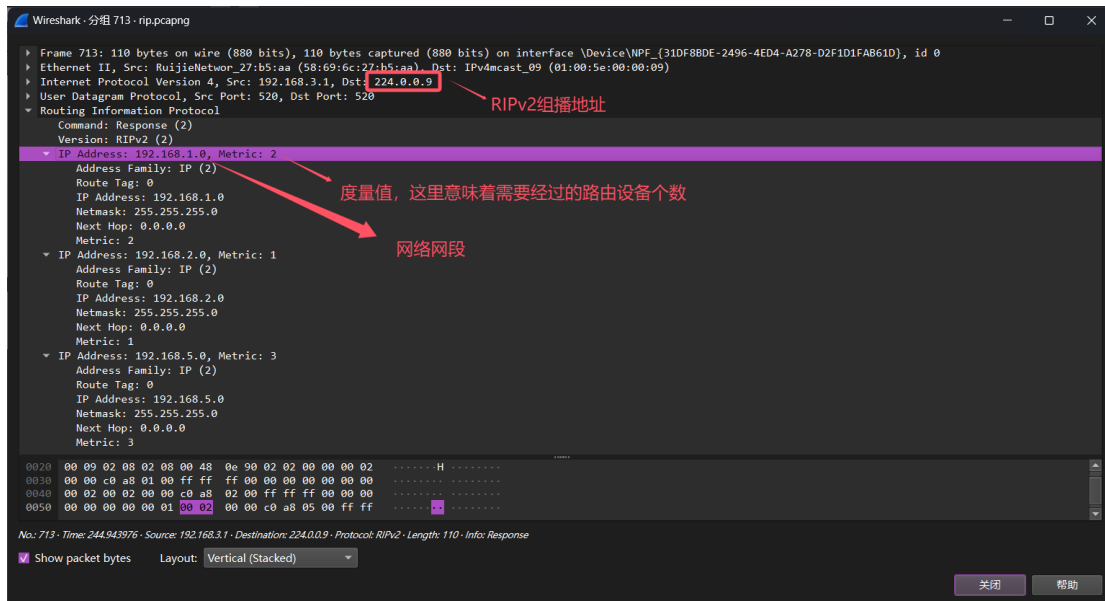
经过测试发现，拔掉网线后，RIP 报文传输明显变快，而当稳定后，报文传输转变成正常的 30s 传输一次。这说明 RIP 会动态感知网络结构，若有变化立即发送路由更新信息通知网络中的路由器，而不必等下一个更新周期到来。这就是 RIP 的触发更新机制。

毒性反转是用来防止出现路由环路（网络故障导致的路由器中相应条目的度量值无限制增大，数据包在路由期间不断传输却无法到达预期网络的现象）的技术。它不会将该网络的路由信息从路由表中立即移除，而是在下一次发送路由更新时，将这个不可达的网络以最大跳数（对于 RIP 来说通常是 15 跳）加 1 的形式广播出去，即设置为 16 跳，表示该网络不可达。

在抓包的 RIP 包中，检查拔掉网线后网络还未收敛时的数据包，发现其度量值为 16，说明出现了毒性反转。



(4) PC1 和 PC2 上均可以捕获到 RIP 包。分析如下：



其中，224.0.0.9 是 v2 的组播地址，在 RIP 报文部分，包含着目的网络的地址，子网掩码，度量值以及最优下一跳地址等信息，其中上图的最优下一跳地址为 0.0.0.0，表示接收这个包的设备需要将最优下一跳设置为发送这个 RIP 包的路由器（这里就是 192.168.3.1）。

实验思考：

(1) 使用 show vlan 命令即可：



```
20-S5750-1(config-if-VLAN 50)#show vlan
VLAN Name                Status    Ports
-----
 1 VLAN0001                STATIC    Gi0/2, Gi0/3, Gi0/4, Gi0/6
                               Gi0/7, Gi0/8, Gi0/9, Gi0/10
                               Gi0/11, Gi0/12, Gi0/13, Gi0/14
                               Gi0/15, Gi0/16, Gi0/17, Gi0/18
                               Gi0/19, Gi0/20, Gi0/21, Gi0/22
                               Gi0/23, Gi0/24, Gi0/25, Gi0/26
                               Gi0/27, Gi0/28
10 VLAN0010                STATIC    Gi0/1
50 VLAN0050                STATIC    Gi0/5
20-S5750-1(config-if-VLAN 50)#ip address 10.10.2.0 255.255.255.0
```

(2) 使用 show ip protocols:

```
20-RSR20-2(config)#show ip protocols
Routing Protocol is "rip"
  Sending updates every 30 seconds
  Invalid after 180 seconds, flushed after 120 seconds
  Outgoing update filter list for all interface is: not set
  Incoming update filter list for all interface is: not set
  Redistribution default metric is 1
  Redistributing:
  Default version control: send version 1, receive version 1
    Interface          Send  Recv
    Serial 2/0         1     1
    GigabitEthernet 0/1 1     1
  Routing for Networks:
    10.0.0.0 255.0.0.0
    192.168.2.0 255.255.255.0
    192.168.3.0 255.255.255.0
  Distance: (default is 120)
  Graceful-restart disabled
```

(3)

RIPv1 的广播地址: 255.255.255.255

RIPv2 的组播地址: 224.0.0.9

(4) 使用 10.10.X.0 的 IP 地址重做本次实验, 注意网段间使用不同的子网掩码。当在 RIPv1 下设置不同网段时, 配置后的端口实际上获得的子网掩码是什么? 配合实验分析原因。

在实验中我们将 PC1 的 IP 地址改为 10.10.0.0 重做本次实验:

```
20-S5750-1(config)#interface vlan 50
20-S5750-1(config-if-VLAN 50)#ip address 10.10.2.0 255.255.0.0
20-S5750-1(config-if-VLAN 50)#no shutdown
20-S5750-1(config-if-VLAN 50)#exit
20-S5750-1(config)#rout
20-S5750-1(config)#route rip
20-S5750-1(config-router)#networ
20-S5750-1(config-router)#network 10.10.0.0
20-S5750-1(config-router)#
```

将版本转换为 RIPv1 后:



```
20-RSR20-2(config-router)#show ip protocols
Routing Protocol is "rip"
  Sending updates every 30 seconds
  Invalid after 180 seconds, flushed after 120 seconds
  Outgoing update filter list for all interface is: not set
  Incoming update filter list for all interface is: not set
  Redistribution default metric is 1
  Redistributing:
  Default version control: send version 1, receive version 1
    Interface          Send  Recv
    Serial 2/0         1     1
    GigabitEthernet 0/1 1     1
  Routing for Networks:
    10.0.0.0 255.0.0.0
    192.168.2.0 255.255.255.0
    192.168.3.0 255.255.255.0
  Distance: (default is 120)
  Graceful-restart disabled
```

用 PC1pingPC2, 可以 ping 通:

The screenshot displays two windows from a network simulation. The top window, titled 'rip', shows the configuration for the Routing Information Protocol. It lists the networks being advertised: 10.0.0.0/24, 192.168.2.0/24, and 192.168.3.0/24, all with a default metric of 1. The bottom window shows the output of a 'tracert' command from PC1 to PC2 (192.168.3.22). The trace shows the path from the source to the destination through three hops, with the final hop being the destination itself. Below the trace, the output of a 'ping' command is shown, indicating that the connection is successful with 4 successful packets and 0% loss.

No.	Time	Source	Destination	Protocol	Length	Info
643	464.069827	192.168.5.1	255.255.255.255	RIPv1	106	Response
660	494.080342	192.168.5.1	255.255.255.255	RIPv1	106	Response
688	524.080724	192.168.5.1	255.255.255.255	RIPv1	106	Response
810	554.080915	192.168.5.1	255.255.255.255	RIPv1	106	Response
831	584.081407	192.168.5.1	255.255.255.255	RIPv1	106	Response
1022	817.669788	192.168.5.1	255.255.255.255	RIPv1	66	Request
1039	818.661293	192.168.5.1	255.255.255.255	RIPv1	66	Request
1046	819.331432	192.168.5.1	255.255.255.255	RIPv1	106	Response
1048	819.660969	192.168.5.1	255.255.255.255	RIPv1	66	Request
1103	849.331760	192.168.5.1	255.255.255.255	RIPv1	106	Response
1114	879.331911	192.168.5.1	255.255.255.255	RIPv1	106	Response
1220	941.815570	10.10.2.0	255.255.255.255	RIPv1	66	Request
1221	942.202884	10.10.2.0	255.255.255.255	RIPv1	106	Response
1222	942.812929	10.10.2.0	255.255.255.255	RIPv1	66	Request
1223	943.812712	10.10.2.0	255.255.255.255	RIPv1	66	Request
1275	972.203260	10.10.2.0	255.255.255.255	RIPv1	106	Response
1329	1002.203635	10.10.2.0	255.255.255.255	RIPv1	106	Response
1359	1032.203859	10.10.2.0	255.255.255.255	RIPv1	106	Response
1373	1062.204383	10.10.2.0	255.255.255.255	RIPv1	106	Response
1405	1092.204697	10.10.2.0	255.255.255.255	RIPv1	106	Response

```
C:\Windows\system32>tracert 192.168.3.22
通过最多 30 个跃点跟踪
到 D52_11 [192.168.3.22] 的路由:
  0  192.168.3.22  0 ms  0 ms  0 ms
  1  10.10.2.0    1 ms  1 ms  1 ms
  2  192.168.1.1  1 ms  1 ms  1 ms
  3  192.168.2.2  41 ms 41 ms 45 ms
  4  D52_11 [192.168.3.22] 43 ms 49 ms 48 ms
跟踪完成。
C:\Windows\system32>ping 192.168.3.22
正在 Ping 192.168.3.22 具有 32 字节的数据:
来自 192.168.3.22 的回复: 字节=32 时间=37ms TTL=125
来自 192.168.3.22 的回复: 字节=32 时间=41ms TTL=125
来自 192.168.3.22 的回复: 字节=32 时间=39ms TTL=125
来自 192.168.3.22 的回复: 字节=32 时间=41ms TTL=125
Ping 统计信息:
    数据包: 已发送 = 4, 已接收 = 4, 丢失 = 0 (0% 丢失),
    往返行程的估计时间(以毫秒为单位):
        最短 = 37ms, 最长 = 41ms, 平均 = 39ms
C:\Windows\system32>
```

实际获得掩码为 255.0.0.0, RIPv1 不支持子网掩码的传递, 默认使用所在网络类别的标准子网掩码。在使用 RIPv1 的网络中, 所有子网都必须使用相同的、基于类别的标准子网掩码, 否则会导致子网掩码信息的不一致和路由错误。

(5) RIPv1 必须使用自动汇总, 不支持不连续网络, 请实验验证。RIPv2 支持不连续网络吗?

自动汇总是指 RIPv1 自动进行路由汇总到其类别地址, 即自动将所有子网汇总到其主类网络地址 (如 A 类、B 类或 C 类)。因为 RIPv1 假设所有子网物理上连续且逻辑上属于同一个网络, 所以 RIPv1 无法正确处理不连续网络。

实验验证:

(1) 网络设置: 在原有的实验拓扑图的结构上, 在两个路由器之间增设一个路由器, 该路由器与两侧路由器相



计算机网络实验报告

连的网段设置成 8 位掩码网络，PC 的网络仍设位 24 位网络（实验室一台机器只有 2 个路由器，我们通过 packet tracer 模拟）

```
C:\Windows\system32>tracert 192.168.3.22

通过最多 30 个跃点跟踪
到 D52_11 [192.168.3.22] 的路由:

  1  <1 毫秒    <1 毫秒    <1 毫秒  192.168.5.1
  2  <1 毫秒    <1 毫秒    <1 毫秒  192.168.1.1
  3  41 ms      41 ms      40 ms    10.10.2.0
  4  49 ms      48 ms      49 ms    D52_11 [192.168.3.22]

跟踪完成。
```

- (2) 配置 RIPv1: 在所有路由器上启用 RIPv1。
- (3) 观察能否 ping 通: 通过观察能否 ping 通验证 RIPv1 必须使用自动汇总，不支持不连续网络。

```
C:\Windows\system32>ping 192.168.3.22

正在 Ping 192.168.3.22 具有 32 字节的数据:
请求超时。
请求超时。
请求超时。
请求超时。

192.168.3.22 的 Ping 统计信息:
    数据包: 已发送 = 4, 已接收 = 0, 丢失 = 4 (100% 丢失),
```

不能 ping 通

总结:

RIPv1 不支持不连续网络的实验预期结果: 由于 RIPv1 进行自动汇总，路由器将无法正确识别不连续网络。上述情况可能导致路由循环或某些网络不可达。

RIPv2 支持不连续网络。RIPv2 允许手动配置路由汇总，并支持子网掩码。

- (6) ripv1 包中没有 tag 位，以下为 ripv2 的报文



计算机网络实验报告

```
Time to Live: 1
Protocol: UDP (17)
Header Checksum: 0x144a [validation disabled]
[Header checksum status: Unverified]
Source Address: 192.168.3.1
Destination Address: 224.0.0.9
[Stream index: 6]
User Datagram Protocol, Src Port: 520, Dst Port: 520
Source Port: 520
Destination Port: 520
Length: 72
Checksum: 0x0e90 [unverified]
[Checksum Status: Unverified]
[Stream index: 4]
[Stream Packet Number: 15]
[Timestamps]
UDP payload (64 bytes)
Routing Information Protocol
Command: Response (2)
Version: RIPv2 (2)
IP Address: 192.168.1.0, Metric: 2
Address Family: IP (2)
Route Tag: 0
IP Address: 192.168.1.0
Netmask: 255.255.255.0
Next Hop: 0.0.0.0
Metric: 2
IP Address: 192.168.2.0, Metric: 1
IP Address: 192.168.5.0, Metric: 3
```

RIPV2 中，有 tag 路由标签，用于在路由策略或过滤中标识特定的路由

Src Port: 源端口是 520，这是 RIP 的标准端口。

Dst Port: 目的端口也是 520。

IP Address 表述该条目所属的 IP 地址，Metric 表示达到该网络的跳数，在 RIP 中，度量值通常表示到达目的网络的跳数

实验内容

实验内容（1）见上述实验。

实验内容（2）

The left screenshot shows a packet capture analysis. It details a UDP packet with source address 192.168.3.1 and destination address 224.0.0.9. The payload is a Routing Information Protocol (RIP) response. It lists three routes: 192.168.1.0 with metric 2, 192.168.2.0 with metric 1, and 192.168.5.0 with metric 3. The right screenshot shows a similar analysis but with a different set of IP addresses and metrics, including 192.168.1.0 with metric 1, 192.168.2.0 with metric 2, and 192.168.3.0 with metric 3.

RIP v1 是一种无类路由协议，它不支持 VLSM，也不支持对网络进行细粒度的子网划分。RIP v1 假设所有子网都使用相同的默认子网掩码，它无法有效支持具有不同子网掩码的网络。

RIP v2 则是有类路由协议，它支持 VLSM，可以在路由信息中携带具体的子网掩码信息，允许不同子网使用不同的子网掩码进行更灵活的路由规划。

可以看到，上图中 netmask 携带了子网掩码的信息



RIP v1 不支持 VLSM，它无法携带子网掩码信息，因此无法为不同的网络分配不同的子网掩码。

RIP v2 支持 VLSM，它能够在路由条目中携带子网掩码，使得同一 RIP 网络中的不同路由可以使用不同的子网掩码。

实验内容（3）

debug ip rip 命令后，显示 debug 信息如下：

```
20-RSR20-1#Nov 20 08:24:41: %SYS-5-CONFIG_I: Configured from console by console
debug ip rip
20-RSR20-1#debug ip rip
20-RSR20-1#Nov 20 08:24:50: %7: [RIP] Update timer expired via interface Serial 2/0[192.168.2.1/24]
*Nov 20 08:24:50: %7: [RIP] Update timer schedule via interface Serial 2/0[192.168.2.1/24]
*Nov 20 08:24:50: %7: [RIP] Prepare to send BROADCAST response...
*Nov 20 08:24:50: %7: [RIP] Building update entries on Serial 2/0
*Nov 20 08:24:50: %7: network 192.168.1.0 metric 1
*Nov 20 08:24:50: %7: network 192.168.5.0 metric 2
*Nov 20 08:24:50: %7: [RIP] Send packet to 192.168.2.255 Port 520 on Serial 2/0
*Nov 20 08:24:53: %7: [RIP] RIP received packet, sock=32979 src=192.168.1.2 len=24
*Nov 20 08:24:53: %7: [RIP] Received version 2 response packet on GigabitEthernet 0/1
*Nov 20 08:24:53: %7: [RIP] Cancel peer[192.168.1.2] remove timer
*Nov 20 08:24:53: %7: [RIP] Peer[192.168.1.2] remove timer schedule ...
*Nov 20 08:24:53: %7: [RIP] Received packet version mismatch
*Nov 20 08:24:57: %7: [RIP] RIP received packet, sock=32979 src=192.168.2.2 len=24
*Nov 20 08:24:57: %7: [RIP] Received version 1 response packet on Serial 2/0
*Nov 20 08:24:57: %7: [RIP] Cancel peer[192.168.2.2] remove timer
*Nov 20 08:24:57: %7: [RIP] Peer[192.168.2.2] remove timer schedule ...
*Nov 20 08:24:57: %7: route-entry: family 2 ip 192.168.3.0 metric 1
*Nov 20 08:24:57: %7: [RIP] Translate mask to 24
*Nov 20 08:24:57: %7: [RIP] [192.168.3.0/24] RIP route update, protocol(4)
*Nov 20 08:24:57: %7: [RIP] Old path is: nhop=192.168.2.2 routesrc=192.168.2.2 intf=2
*Nov 20 08:24:57: %7: [RIP] New path is: nhop=192.168.2.2 routesrc=192.168.2.2 intf=2
*Nov 20 08:24:57: %7: [RIP] [192.168.3.0/24] RIP distance apply from 192.168.2.2!
*Nov 20 08:24:57: %7: [RIP] [192.168.3.0/24] cancel Route timer
*Nov 20 08:24:57: %7: [RIP] [192.168.3.0/24] route timer schedule ...
```

这是 RIPv1 的调试信息，从调试信息中，我们可以知道整个 RIP 学习的具体流程，其中展示了一个错误，显示收到的 RIP 包版本不匹配（在 0/1 端口接收到 RIPv2 数据包），由此我们顺藤摸瓜，找出了交换机上信息配置不正确，成功解决了错误。

使用 debug ip packet 命令，显示 debug 信息如下：

```
20-RSR20-2#debug ip packet
20-RSR20-2#Nov 20 08:49:18: %7: IP: s=192.168.3.22 (GigabitEthernet 0/1), d=192.168.3.255, vrf=global(0), len=1468, received
debug ip packet: Nov 20 08:49:19: %7: IP: s=192.168.2.1 (Serial 2/0), d=255.255.255.255, vrf=global(0), len=72, received
*Nov 20 08:49:25: %7: IP: s=192.168.2.2 (Local), d=255.255.255.255 (Serial 2/0), vrf=global(0), len=52, sent ip pkt to link_layer 222
*Nov 20 08:49:27: %7: IP: s=192.168.3.22 (GigabitEthernet 0/1), d=192.168.3.255, vrf=global(0), len=1468, received
*Nov 20 08:49:32: %7: IP: s=192.168.3.1 (Local), d=255.255.255.255 (GigabitEthernet 0/1), vrf=global(0), len=1468, received
*Nov 20 08:49:35: %7: IP: s=192.168.3.22 (GigabitEthernet 0/1), d=192.168.3.255, vrf=global(0), len=1468, received
*Nov 20 08:49:44: %7: IP: s=192.168.3.22 (GigabitEthernet 0/1), d=192.168.3.255, vrf=global(0), len=1468, received
*Nov 20 08:49:49: %7: IP: s=192.168.2.1 (Serial 2/0), d=255.255.255.255, vrf=global(0), len=72, received
*Nov 20 08:49:52: %7: IP: s=192.168.3.22 (GigabitEthernet 0/1), d=192.168.3.255, vrf=global(0), len=1468, received
*Nov 20 08:49:55: %7: IP: s=192.168.2.2 (Local), d=255.255.255.255 (Serial 2/0), vrf=global(0), len=52, sent ip pkt to link_layer 222
*Nov 20 08:50:01: %7: IP: s=192.168.3.22 (GigabitEthernet 0/1), d=192.168.3.255, vrf=global(0), len=1468, received
*Nov 20 08:50:02: %7: IP: s=192.168.3.1 (Local), d=255.255.255.255 (GigabitEthernet 0/1), vrf=global(0), len=1468, received
*Nov 20 08:50:09: %7: IP: s=192.168.3.22 (GigabitEthernet 0/1), d=192.168.3.255, vrf=global(0), len=1468, received
*Nov 20 08:50:18: %7: IP: s=192.168.3.22 (GigabitEthernet 0/1), d=192.168.3.255, vrf=global(0), len=1468, received
*Nov 20 08:50:19: %7: IP: s=192.168.2.1 (Serial 2/0), d=255.255.255.255, vrf=global(0), len=72, received
*Nov 20 08:50:25: %7: IP: s=192.168.2.2 (Local), d=255.255.255.255 (Serial 2/0), vrf=global(0), len=52, sent ip pkt to link_layer 222
*Nov 20 08:50:26: %7: IP: s=192.168.3.22 (GigabitEthernet 0/1), d=192.168.3.255, vrf=global(0), len=1468, received
*Nov 20 08:50:32: %7: IP: s=192.168.3.1 (Local), d=255.255.255.255 (GigabitEthernet 0/1), vrf=global(0), len=1468, received
*Nov 20 08:50:35: %7: IP: s=192.168.3.22 (GigabitEthernet 0/1), d=192.168.3.255, vrf=global(0), len=1468, received
*Nov 20 08:50:43: %7: IP: s=192.168.3.22 (GigabitEthernet 0/1), d=192.168.3.255, vrf=global(0), len=1468, received
*Nov 20 08:50:49: %7: IP: s=192.168.2.1 (Serial 2/0), d=255.255.255.255, vrf=global(0), len=72, received
*Nov 20 08:50:52: %7: IP: s=192.168.3.22 (GigabitEthernet 0/1), d=192.168.3.255, vrf=global(0), len=1468, received
*Nov 20 08:50:55: %7: IP: s=192.168.2.2 (Local), d=255.255.255.255 (Serial 2/0), vrf=global(0), len=52, sent ip pkt to link_layer 222
*Nov 20 08:51:01: %7: IP: s=192.168.3.22 (GigabitEthernet 0/1), d=192.168.3.255, vrf=global(0), len=1468, received
*Nov 20 08:51:02: %7: IP: s=192.168.3.1 (Local), d=255.255.255.255 (GigabitEthernet 0/1), vrf=global(0), len=1468, received
*Nov 20 08:51:09: %7: IP: s=192.168.3.22 (GigabitEthernet 0/1), d=192.168.3.255, vrf=global(0), len=1468, received
*Nov 20 08:51:18: %7: IP: s=192.168.3.22 (GigabitEthernet 0/1), d=192.168.3.255, vrf=global(0), len=1468, received
*Nov 20 08:51:19: %7: IP: s=192.168.2.1 (Serial 2/0), d=255.255.255.255, vrf=global(0), len=72, received
*Nov 20 08:51:25: %7: IP: s=192.168.3.22 (GigabitEthernet 0/1), d=192.168.3.255, vrf=global(0), len=1468, received
*Nov 20 08:51:32: %7: IP: s=192.168.3.1 (Local), d=255.255.255.255 (GigabitEthernet 0/1), vrf=global(0), len=1468, received
*Nov 20 08:51:35: %7: IP: s=192.168.3.22 (GigabitEthernet 0/1), d=192.168.3.255, vrf=global(0), len=1468, received
*Nov 20 08:51:43: %7: IP: s=192.168.3.22 (GigabitEthernet 0/1), d=192.168.3.255, vrf=global(0), len=1468, received
*Nov 20 08:51:49: %7: IP: s=192.168.2.1 (Serial 2/0), d=255.255.255.255, vrf=global(0), len=72, received
*Nov 20 08:51:52: %7: IP: s=192.168.3.22 (GigabitEthernet 0/1), d=192.168.3.255, vrf=global(0), len=1468, received
```

从中我们可以获取到 rip 数据包的传输信息。比如 RIPv1 的广播地址为 255.255.255.255

实验内容 4



```
*Nov 20 09:00:42: %7: [RIP] Update timer expired via interface GigabitEthernet 0/1[192.168.1.1/24]
*Nov 20 09:00:42: %7: [RIP] Update timer schedule via interface GigabitEthernet 0/1[192.168.1.1/24]
*Nov 20 09:00:42: %7: [RIP] Prepare to send BROADCAST response...
*Nov 20 09:00:42: %7: [RIP] Building update entries on GigabitEthernet 0/1
network 192.168.2.0 metric 1
network 192.168.3.0 metric 2
*Nov 20 09:00:42: %7: [RIP] Send packet to 192.168.1.255 Port 520 on GigabitEthernet 0/1
*Nov 20 09:00:50: %7: [RIP] RIP received packet, sock=32979 src=192.168.2.2 len=24
*Nov 20 09:00:50: %7: [RIP] Received version 1 response packet on Serial 2/0
*Nov 20 09:00:50: %7: [RIP] Cancel peer[192.168.2.2] remove timer
*Nov 20 09:00:50: %7: [RIP] Peer[192.168.2.2] remove timer schedule...
*Nov 20 09:00:50: %7: [RIP] route-entry: family 2 ip 192.168.3.0 metric 16
*Nov 20 09:00:50: %7: [RIP] Translate mask to 24
*Nov 20 09:00:50: %7: [RIP] [192.168.3.0/24] RIP route disabling...
*Nov 20 09:00:50: %7: [RIP] [192.168.3.0/24] cancel route timer
*Nov 20 09:00:50: %7: [RIP] [192.168.3.0/24] route timer schedule...
*Nov 20 09:00:50: %7: [RIP] Trigger timer Schedule, by instance 0
*Nov 20 09:00:50: %7: [RIP] [192.168.3.0/24] ready to add into kernel...
*Nov 20 09:00:50: %7: [RIP] NSM delete: IPv4 Route 192.168.3.0/24
*Nov 20 09:00:50: %7: [RIP] Update timer schedule via interface Serial 2/0[192.168.2.1/24]
*Nov 20 09:00:50: %7: [RIP] Update timer schedule via interface Serial 2/0[192.168.2.1/24]
*Nov 20 09:00:50: %7: [RIP] Prepare to send BROADCAST response...
*Nov 20 09:00:50: %7: [RIP] Building update entries on Serial 2/0
network 192.168.1.0 metric 1
network 192.168.5.0 metric 2
*Nov 20 09:00:50: %7: [RIP] Send packet to 192.168.2.255 Port 520 on Serial 2/0
*Nov 20 09:00:53: %7: [RIP] RIP received packet, sock=32979 src=192.168.1.2 len=24
*Nov 20 09:00:53: %7: [RIP] Received version 1 response packet on GigabitEthernet 0/1
*Nov 20 09:00:53: %7: [RIP] Cancel peer[192.168.1.2] remove timer
*Nov 20 09:00:53: %7: [RIP] Peer[192.168.1.2] remove timer schedule...
*Nov 20 09:00:53: %7: [RIP] route-entry: family 2 ip 192.168.5.0 metric 1
*Nov 20 09:00:53: %7: [RIP] Translate mask to 24
*Nov 20 09:00:53: %7: [RIP] [192.168.5.0/24] RIP route update, protocol(4)
*Nov 20 09:00:53: %7: [RIP] Old path is: nhop=192.168.1.2 routesrc=192.168.1.2 intf=5
*Nov 20 09:00:53: %7: [RIP] New path is: nhop=192.168.1.2 routesrc=192.168.1.2 intf=5
*Nov 20 09:00:53: %7: [RIP] [192.168.5.0/24] RIP distance apply from 192.168.1.2!
*Nov 20 09:00:53: %7: [RIP] [192.168.5.0/24] cancel Route timer
*Nov 20 09:00:53: %7: [RIP] [192.168.5.0/24] route timer schedule...
*Nov 20 09:00:53: %7: [RIP] Trigger timer expired, by instance 0
*Nov 20 09:00:54: %7: [RIP] Prepare to send BROADCAST response...
*Nov 20 09:00:54: %7: [RIP] Building update entries on Serial 2/0
*Nov 20 09:00:54: %7: [RIP] Skip route[192.168.1.0/24] in trigger
*Nov 20 09:00:54: %7: [RIP] Skip route[192.168.2.0/24] in trigger
*Nov 20 09:00:54: %7: [RIP] Skip route[192.168.5.0/24] in trigger
*Nov 20 09:00:54: %7: [RIP] Skip send response packet...
*Nov 20 09:00:54: %7: [RIP] Prepare to send BROADCAST response...
*Nov 20 09:00:54: %7: [RIP] Building update entries on GigabitEthernet 0/1
*Nov 20 09:00:54: %7: [RIP] Skip route[192.168.1.0/24] in trigger
*Nov 20 09:00:54: %7: [RIP] Skip route[192.168.2.0/24] in trigger
*Nov 20 09:00:54: %7: [RIP] network 192.168.3.0 metric 16
*Nov 20 09:00:54: %7: [RIP] Skip route[192.168.5.0/24] in trigger
*Nov 20 09:00:54: %7: [RIP] Send packet to 192.168.1.255 Port 520 on GigabitEthernet 0/1
*Nov 20 09:00:57: %7: [RIP] RIP received packet, sock=32979 src=192.168.2.2 len=24
*Nov 20 09:00:57: %7: [RIP] Received version 1 response packet on Serial 2/0
*Nov 20 09:00:57: %7: [RIP] Cancel peer[192.168.2.2] remove timer
*Nov 20 09:00:57: %7: [RIP] Peer[192.168.2.2] remove timer schedule...
*Nov 20 09:00:57: %7: [RIP] route-entry: family 2 ip 192.168.3.0 metric 16
*Nov 20 09:00:57: %7: [RIP] Translate mask to 24
```

观察图中 metric 相关参数（16 变 1 变 16），可以发现链路间的变化
在这个过程中，路由表中的 ip 信息发生交换

在链路断开前，路由表如下图所示

```
show ip route

Codes: C - connected, S - static, R - RIP, B - BGP
       O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default

Gateway of last resort is no set
C    192.168.1.0/24 is directly connected, GigabitEthernet 0/1
C    192.168.1.1/32 is local host.
C    192.168.2.0/24 is directly connected, Serial 2/0
C    192.168.2.1/32 is local host.
R    192.168.5.0/24 [120/1] via 192.168.1.2, 00:18:38, GigabitEthernet 0/1
20-BSR20-1#
```

断开之后，如下图，可以看到路由表新增了 192.168.3.0 的信息
度量值 120 暗示这是 RIP 协议，后接 1 表示跳数，因为 RIP 的默认度量值是基于跳数的



```
show ip route
```

```
Codes: C - connected, S - static, R - RIP, B - BGP  
O - OSPF, IA - OSPF inter area  
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2  
E1 - OSPF external type 1, E2 - OSPF external type 2  
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2  
ia - IS-IS inter area, * - candidate default
```

```
Gateway of last resort is no set
```

```
C 192.168.1.0/24 is directly connected, GigabitEthernet 0/1  
C 192.168.1.1/32 is local host.  
C 192.168.2.0/24 is directly connected, Serial 2/0  
C 192.168.2.1/32 is local host.  
R 192.168.3.0/24 [120/1] via 192.168.2.2, 00:00:06, Serial 2/0  
R 192.168.5.0/24 [120/1] via 192.168.1.2, 00:19:54, GigabitEthernet 0/1
```